

ECE 622: Modeling & Verification of Embedded Systems

Spring 2026: Lab 3: Hybrid Systems Verification

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Overview

In this lab we used **PHAVerLite** to compute forward reachable sets of hybrid automata and to check safety-style properties. For each problem we encoded locations with continuous dynamics (written in PHAVer syntax using primed variables, e.g. $\mathbf{x}' == \mathbf{v}$ for $\dot{x} = v$) and discrete jumps with guards/resets, then bounded the analysis using reasonable location invariants. We varied the partition constraint (**pc**) to study the tradeoff between runtime and precision: smaller **pc** refines the polyhedral over-approximation and can remove spurious reachability, but increases the number of symbolic locations and CPU time. Throughout, we treated results in the standard soundness direction for over-approximation [Frehse, 2008]: “**not reachable**” for a bad set is a proof for the modeled system, whereas “**reachable**” indicates the property is not proven at that abstraction level and may require refinement. To keep experiments reproducible, long runs were capped with a fixed timeout and recorded as timed-out when necessary.

1 Part A: Analysis of Bouncing Ball Model

The bouncing-ball experiment is the canonical hybrid model (continuous flight with gravity, discrete bounce resets). The lab model instantiates the template from Canvas for reachability experiments below.

A.1: Reachable sets for two values of **pc**

Use PHAVerLite to compute reachable states with $\mathbf{pc}=0.5$ and $\mathbf{pc}=0.05$. Plot both on the same axes (clearly distinguishable). Answer: for each $\mathbf{pc}$, is $(x, v) = (0.55, 0.0)$ unreachable based on the plot?

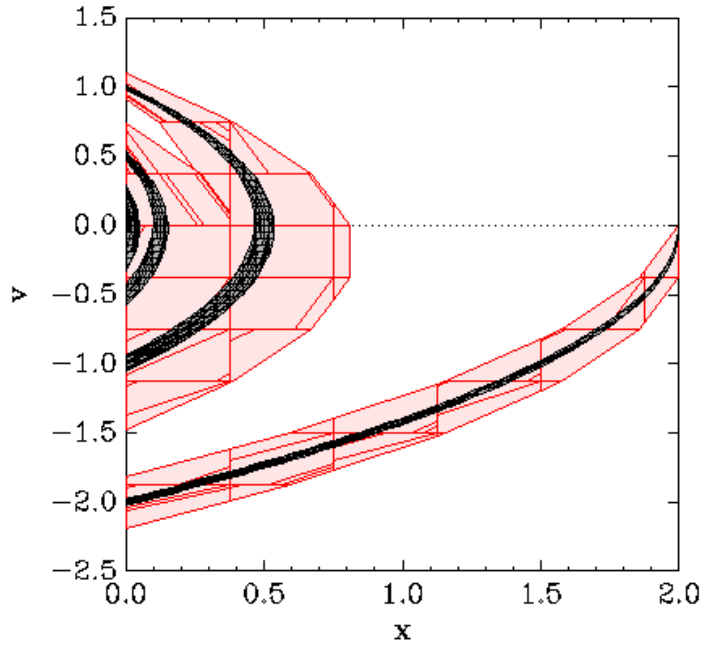


Figure 1: A.1: Reachable-state plot used to classify the target $(x, v) = (0.55, 0.0)$ at two partition settings.

We first used the reachable-set plot as a coarse check of whether the target point is plausibly excluded by the reachable enclosure. At the coarser partition ($pc=0.5$), the enclosure is visibly wider and the target point is not excluded by the plot. After refining to $pc=0.05$, the enclosure tightens and the target point is visually outside the reachable region.

For A.1, the plot at $pc=0.5$ does not let us conclude unreachability of $(0.55, 0)$, while the refined plot at $pc=0.05$ places the point outside the reachable set.

Summary: With a coarse partition ($pc=0.5$), the target is covered by the reachable set, so exclusion cannot be claimed. With the finer partition ($pc=0.05$), the target point is outside the reachable set, and we can visually conclude it is unreachable at this precision.

A.2: `is_reachable` vs. `plot`; sweep `pc`

Verify $(0.55, 0.0)$ with `is_reachable` for $pc=0.5$ and 0.05 (consistency with A.1). Then sweep `pc` over 0.5, 0.2, 0.1, 0.05, 0.01, and 0.005. If any run exceeds 15 minutes, report time as “timed out.”

Command form for reachability check:

```
target = bball.{falling & x==55/100 & v==0};
check = bball.is_reachable(target);
```

Partition pc	Condition reachable?	Num locations reached	CPU time (s)
0.5	yes	33	0.03
0.2	yes	71	0.077
0.1	yes	152	0.171
0.05	no	327	0.439
0.01	no	2671	8.814
0.005	no	5373	44.513

Table 1: A.2 reachability sweep from PHAVerLite outputs in `src/lab3.html`.

The sweep shows the expected precision-runtime tradeoff: as pc decreases, the number of symbolic locations and CPU time increase sharply, but the reachable enclosure becomes tighter. In our runs, reachability of state comes to a conclusion between $pc=0.1$ (reachable) and $pc=0.05$ (not reachable), which is consistent with refinement removing spurious reachability.

For the follow-up in A.2, among $pc=1.0$, 0.075 , and 0.001 , the finest setting $pc=0.001$ is the one most likely to report the target as unreachable.

A.3: Bounce counter n and peak-height queries

In A.3 we used two complementary checks: a geometric inspection of the reachable enclosure to see whether $(0.15, 0)$ lies on a plausible peak, and a direct reachability query indexed by bounce count. Both checks point to the same conclusion: the second-bounce peak is consistent with height 0.15 at zero velocity, while the other peaks do not appear supported at this partition.

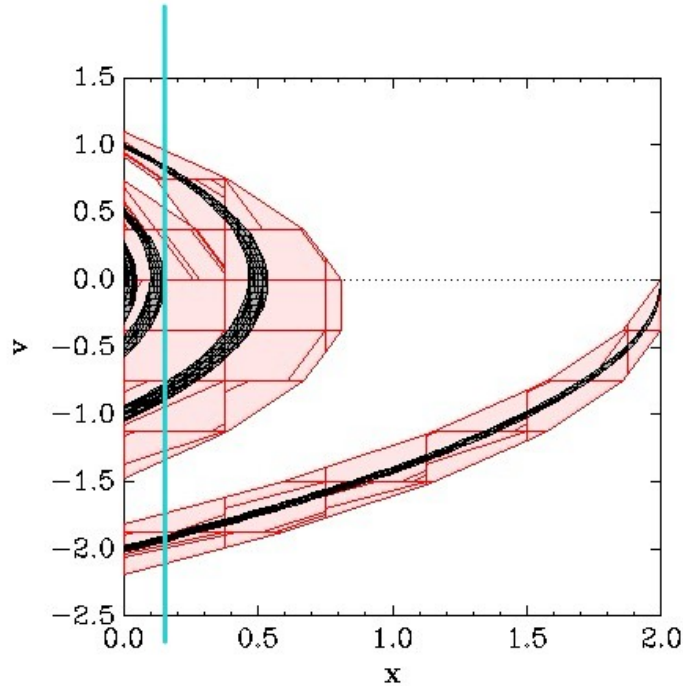


Figure 2: A.3: Reachable-set plot used to inspect whether $(x, v) = (0.15, 0)$ aligns with bounce peaks.

```
check = bball.is_reachable(n1);
check = bball.is_reachable(n2);
check = bball.is_reachable(n3);
```

```
check = bball.is_reachable(n4);
```

```
darling@jkq:~$ ./phaverlite bouncing_ball.pha
PHAVerLite 0.7: starting
  Parsing file bouncing_ball.pha
  Computing reachable states of bball
  Time in get_reach_set: 0.629 secs
n1 not reachable.
  Computing reachable states of bball
  Time in get_reach_set: 0.162 secs
n2 is reachable.
  Computing reachable states of bball
  Time in get_reach_set: 0.203 secs
n3 not reachable.
  Computing reachable states of bball
  Time in get_reach_set: 0.187 secs
n4 not reachable.
  Time in file bouncing_ball.pha: 1.190 secs
Time in PHAVerLite: 1.191 secs
```

Figure 3: A.3: Terminal output for bounce-indexed queries **n1–n4**.

In A.3, both the plot and the bounce-indexed reachability checks indicate that height 0.15 at zero velocity is consistent with the **second bounce** at $pc=0.05$.

2 Part B: Model of Heating System

The heating plant is a two-location hybrid: temperature evolves continuously in each mode while a clock t advances at rate one, so time-bounded properties can be expressed as conjunctions involving t . Location invariants (e.g. $0 \leq t \leq 10$) keep the reachable set compact for PHAVerLite.

B.1: Reachable set with $pc=0.3$

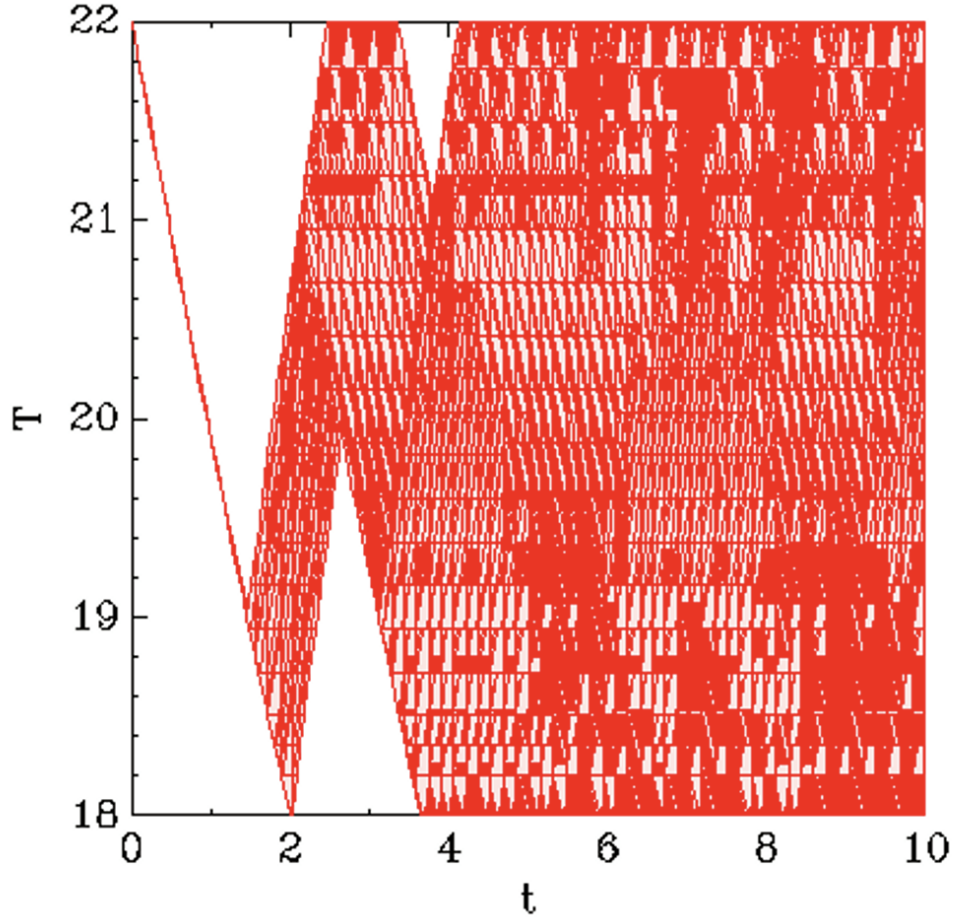


Figure 4: B.1: Heating-system reachable set (time on x -axis, temperature on y -axis) at $pc=0.3$.

We implemented the clocked hybrid model and plotted the reachable set in the time–temperature plane. The resulting tube provides a compact summary of the system’s possible temperature evolution over the bounded horizon, and it makes the effect of switching between modes visible as changes in the reachable enclosure. In B.1, we generated the time–temperature reachable set over $t \in [0, 10]$ using $pc=0.3$.

B.2: Time-bounded temperature property

This experiment models a heating system starting in the *cool* location with clock $t = 0$ and temperature $x = 22$. We are to check: is it ever possible for $x < 20$ within $t = 0.98$ seconds? We encode the “bad” set as the slice $t = 0.98$ and $x < 20$, and use `is_reachable` to see if this state is attainable for several values of the partition constraint (pc).

```
bad = heatingsystem.{ $ & t == 98/100 & x < 20 };  
check = heatingsystem.is_reachable(bad);
```

Sweeping across pc , our results show that for $pc=1.0$ and $pc=1.2$, the bad set remains **not reachable**, but when $pc=1.2$ or larger, the over-approximation becomes too coarse and the bad set is reported as **reachable**. Thus, the largest value of pc (within 0.1 of the true threshold) for which the bad set is not reachable is $pc=1.2$. **Result:** The largest partition constraint for which the system reliably proves the property (unreachability of the bad set) is $pc=1.2$; for $pc \geq 1.2$, the abstraction is too coarse to exclude $x < 20$ at $t = 0.98$.

We expressed the requirement as a bad set at the exact time slice $t = 0.98$ and then swept pc to find where the abstraction becomes too coarse to exclude the bad region. In our sweep, the check remained *not reachable* below $pc=1.3$ and became *reachable* at $pc=1.3$, giving a tight bracket for the requested threshold.

```
bad = heatingsystem.{ $ & t == 98/100 & x > 20 };
check = heatingsystem.is_reachable(bad);
```

In B.2, the largest partition constraint that still reports the bad set as **not reachable** is $\text{pc}=1.2$; at $\text{pc}=1.3$, the bad set becomes **reachable**.

3 Part C: Car Acceleration and Braking

Scenario: two intersections 500 m apart; stop sign then traffic light. Full stop at sign; accelerate at 3.5 m s^{-2} . Braking for red light starts between 250 m and 300 m past the stop sign (i.e. 200 m–250 m from the light). Braking acceleration -5.5 m s^{-2} until stop.

The red-light onset is modeled as **nondeterministic**: the jump from acceleration to braking may occur at any position in $[250 \text{ m}, 300 \text{ m}]$. The invariant $x \leq 300 \text{ m}$ in the accel location forces a decision before leaving that window.

C.1: Hybrid automaton diagram

Circle locations; label guards on jumps; derivatives of position and velocity per location; location invariants (including to force transitions when guards stay true).

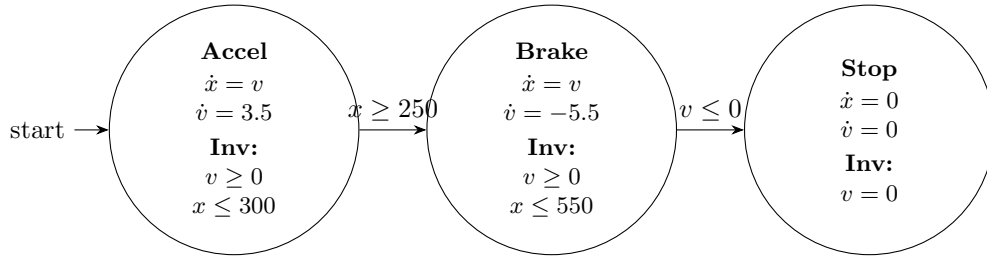


Figure 5: C.1: Hybrid automaton for the car (initial full stop at $x = 0$).

Quickly checking using equations of motion

Uniform acceleration segments satisfy the usual SUVAT relations; they are useful to sanity-check extreme speeds and stopping distances against the reachable set, not as a substitute for PHAVerLite proofs. The equations of motion are:

$$v^2 = u^2 + 2as, \quad (1)$$

$$s = ut + \frac{1}{2}at^2, \quad (2)$$

$$v = u + at. \quad (3)$$

We used these equations to compute simple bounds that explain the reachable envelopes. Starting from rest ($u = 0$) with $a = 3.5 \text{ m s}^{-2}$, the speed at the braking window endpoints is:

$$v(250) = \sqrt{2 \cdot 3.5 \cdot 250} \approx 41.8 \text{ m s}^{-1}, \quad v(300) = \sqrt{2 \cdot 3.5 \cdot 300} \approx 45.8 \text{ m s}^{-1}.$$

The corresponding times to reach those positions follow from (2) with $u = 0$:

$$t(250) = \sqrt{\frac{2 \cdot 250}{3.5}} \approx 11.95 \text{ s}, \quad t(300) = \sqrt{\frac{2 \cdot 300}{3.5}} \approx 13.09 \text{ s}.$$

Once braking begins ($a = -5.5 \text{ m s}^{-2}$), (3) shows velocity decreases monotonically, so the maximum braking speed is the entry speed at the braking trigger. This is why Property 2 is expected to hold for the concrete dynamics, while Property 3 can fail depending on when braking is triggered (late braking can still enter at speeds above 40 m s^{-1}).

C.2: PHAVer model and reachable set

Translate to PHAVer. Use reasonable invariants to bound the state space. With $pc=3.0$, plot reachable states: position (x -axis) vs. velocity (y -axis).

We encoded the car dynamics and red-light nondeterminism in PHAVerLite, bounded the state space with simple invariants, and generated the reachable set at $pc=3.0$. We then exported the reachable region and plotted position versus velocity to visualize the envelope of possible behaviors under the uncertain braking start.

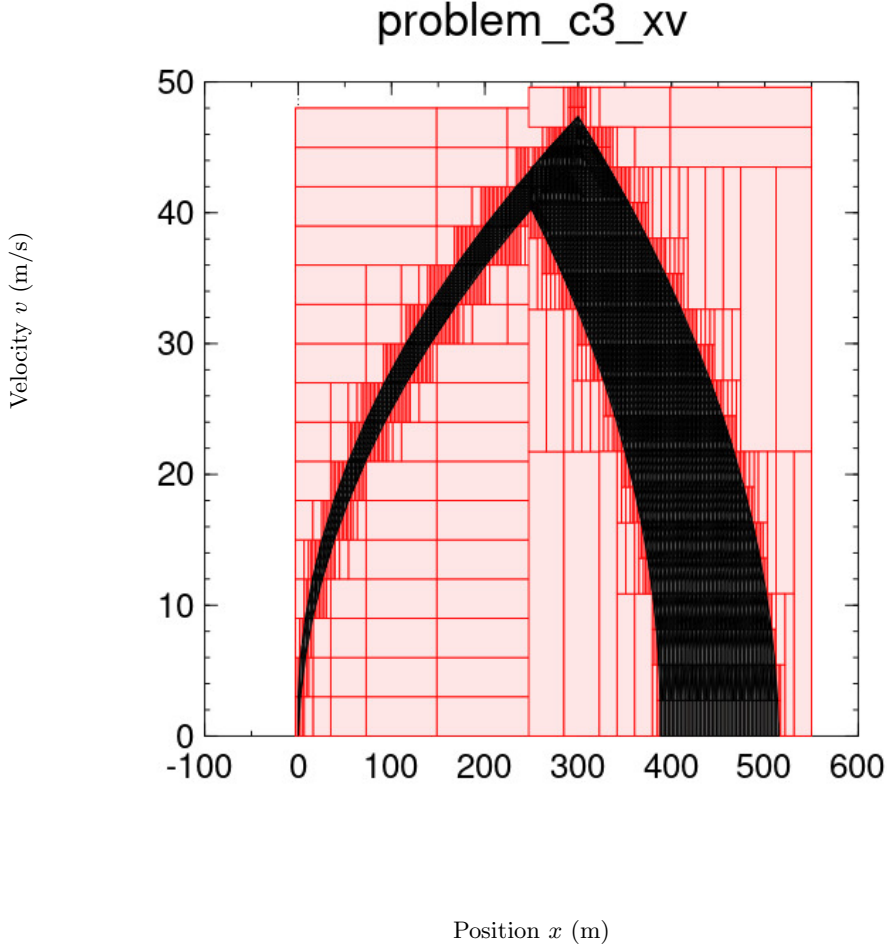


Figure 6: C.2: Reachable states in the (x, v) plane at $pc=3.0$ (export from PHAVerLite), with axis labels for position and velocity.

In C.2, we generated the reachable states in the (x, v) plane at $pc=3.0$, with position on the horizontal axis and velocity on the vertical axis.

C.3: Three reachability properties

Property statement (compact)	$pc=3.0$	$pc=1.2$
Stops before the second intersection	not proven	satisfied
$v \leq 46.5 \text{ m s}^{-1}$ during braking	not proven	satisfied
$v \leq 40.0 \text{ m s}^{-1}$ during braking	not proven	not proven

Table 2: C.3 property statements and outcomes at two partition constraints.

Bad-state queries (submitted jkhare_C3.pha). Each property is checked by reachability of a *violation* set; proving the property would require `not reachable` for that set.

```
p1_bad = car_sys.{Brake & x >= 500 & v >= 0.001};
p1_bad_reach = car_sys.is_reachable(p1_bad);
p2_bad = car_sys.{Brake & v >= 46.5};
p2_bad_reach = car_sys.is_reachable(p2_bad);
p3_bad = car_sys.{Brake & v >= 40.0};
p3_bad_reach = car_sys.is_reachable(p3_bad);
```

Listing 1: C.3: PHAVerLite output summary for the three property checks with `pc=3.0`

```
p1_bad is reachable.
p2_bad is reachable.
p3_bad is reachable.
```

Here $v \geq 0.001$ in P1 separates “still moving” from a full stop at $v = 0$. At `pc=3.0`, PHAVerLite reported **reachable** for all three bad sets; therefore **none** of the three properties is formally proven as stated (unreachability would be required for a sound proof). Reachability here may be due to over-approximation, but it does not establish safety. A tighter `pc` would be needed to refine our conclusion.

To distinguish genuine violations from conservatism, we reran the same C.3 checks with a tighter partition (`pc=1.2`). In that refined run, PHAVerLite reported Property 1 and Property 2 bad sets as **not reachable**, while Property 3 remained reachable.

Listing 2: C.3: PHAVerLite output summary for the three property checks with `pc=1.2`

```
p1_bad is not reachable.
p2_bad is not reachable.
p3_bad is reachable.
```

Verifying Properties from Physics

P1: Stops before the second intersection

We verify Property 1 using physics. Let $x_b \in [250, 300]$ be the braking-trigger position. From rest in the acceleration phase,

$$v_b^2 = 2a_{acc}x_b = 2(3.5)x_b.$$

In the braking phase with $a_{br} = -5.5$, the stopping distance is

$$d_{stop} = \frac{v_b^2}{2 \cdot 5.5}.$$

So the stop position is

$$x_{stop} = x_b + d_{stop} = x_b + \frac{2 \cdot 3.5 x_b}{2 \cdot 5.5} = x_b \left(1 + \frac{7}{11}\right) = \frac{18}{11}x_b.$$

Worst case is $x_b = 300$, giving

$$x_{stop, max} = \frac{18}{11} \cdot 300 \approx 490.9 \text{ m} < 500 \text{ m}.$$

Thus, the concrete dynamics satisfy Property 1. This agrees with the refined PHAVer result (`pc=1.2`: bad set not reachable) and explains why the coarse `pc=3.0` run can be conservative.

P2: $v \leq 46.5 \text{ ms}^{-1}$ during braking

We verify Property 2 using physics. The car starts from rest and accelerates at 3.5 ms^{-2} for at most 300 m, so the largest possible entry speed into braking is

$$v_{max} = \sqrt{2as} = \sqrt{2 \cdot 3.5 \cdot 300} \approx 45.8 \text{ ms}^{-1} < 46.5 \text{ ms}^{-1}.$$

Since braking only decreases speed, Property 2 is satisfied by the concrete dynamics. This is consistent with the refined PHAVer run (`pc=1.2`: bad set not reachable), while the coarse `pc=3.0` verdict is conservatively inconclusive.

P3: $v \leq 40.0 \text{ m s}^{-1}$ during braking

We verify Property 3 using physics. Braking can begin anywhere in $x_b \in [250, 300]$, so the braking-entry speed range is

$$v_b = \sqrt{2 \cdot 3.5 x_b} \in \left[\sqrt{2 \cdot 3.5 \cdot 250}, \sqrt{2 \cdot 3.5 \cdot 300} \right] \approx [41.8 \text{ m s}^{-1}, 45.8 \text{ m s}^{-1}].$$

Because the minimum possible entry speed already exceeds 40 m s^{-1} , the violation condition for Property 3 is physically possible (indeed immediately at braking entry). So Property 3 is not valid for the concrete model, matching PHAVer reachability at both $\text{pc}=3.0$ and $\text{pc}=1.2$.

In C.3, the coarse run at $\text{pc}=3.0$ does not prove any of the three claims, while the refined run at $\text{pc}=1.2$ proves Property 1 and Property 2 (bad sets not reachable) but still does not prove Property 3.

C.4: 0–60 MPH car speed estimate

Add a time variable; $60 \text{ mph} = 26.8 \text{ m s}^{-1}$. Use $\text{pc}=3.0$. Two queries: (i) 26.8 m s^{-1} unreachable by time T ; (ii) reachable by time $T + 0.1 \text{ s}$. Report an interval for 0–60 time. Submit `<lastname>_C4.pha`.

A separate automaton `car_c4` adds t with $\dot{t} = 1$ in the `Accel` location (constant acceleration from rest, $\dot{v} = 3.5$). Bracketing queries at $T = 7.6 \text{ s}$ and $T + 0.1 \text{ s} = 7.7 \text{ s}$:

```
q_before = car_c4.{Accel & t <= 7.6 & v >= 26.8};
r_before = car_c4.is_reachable(q_before);
q_after = car_c4.{Accel & t <= 7.7 & v >= 26.8};
r_after = car_c4.is_reachable(q_after);
```

With $\text{pc}=3.0$, the tool reported `q_before` not reachable and `q_after` reachable, so the 0–60 mph time lies in the interval $(7.6 \text{ s}, 7.7 \text{ s}]$ (consistent with $t = v/a$ for constant a at 26.8 m s^{-1}).

In C.4, the model brackets the 0–60 mph time in the interval $(7.6 \text{ s}, 7.7 \text{ s}]$.

Conclusion

Across Parts A and B we saw the central practical lesson of PHAVerLite reachability: decreasing pc substantially increases cost (locations and runtime) but can sharpen the reachable enclosure enough to change the reachability verdict for a boundary target. For the heating model, sharpening provided a concrete way to locate where the abstraction stops being strong enough to exclude a time-slice counterexample.

In Part C, we encoded environmental uncertainty as nondeterminism in the braking trigger and checked properties as bad-state reachability queries. In our runs, the three C.3 bad sets were reachable at $\text{pc}=3.0$ (hence not proven, or not proved "unprovable"). For C.4 bracketing approach produced a narrow and interpretable 0–60 mph time interval, so we just which could tell us some interesting facts about how this the roadside "STOP" indicator placement could be optimized considering the various car models we created for based on performance too!

Appendix: Tooling and submission artifacts

This report template was adapted from the Lab 2 `latex-build` layout. Build with `latexmk` (see `README.md`). PHAVerLite models and plots were produced on the course toolchain; figures can be synced via Git alongside this `.tex` file.

Reproducibility (Part C). Scripts under `lab3/modeling-car/`: `run_c3.sh` (timeout-wrapped PHAVerLite), `check_c3_properties.sh`, `plot_c3.sh` for the (x, v) figure; `lab3/modeling-car/c4/run_c4.sh` for C.4 bracketing runs.

Files submitted:

- `SHIH_A3.pha`
- `SHIH_B2.pha`
- `jkhare_C3.pha`
- `jkhare_C4.pha`

References

Goran Frehse. PHAVer: Algorithmic verification of hybrid systems past HyTech. *International Journal on Software Tools for Technology Transfer*, 10(3):263–279, 2008. doi: 10.1007/s10009-007-0062-3.